

Building the Chicago ENCOUNTER Network (nurse-run Community Effective primary care)

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Background. Chicago ENCOUNTER is a equity-focused healthcare effort between Illinois Tech and Rush University to reimagine the infrastructure for primary care in Chicago. Driven by persistent and deadly deficits in human health among residents of Chicago’s most under-invested neighborhoods [7, 8], the ENCOUNTER project seeks to define a feasible strategy for promoting equitable access to, and utilization of, primary care. Primary care is essential to equity, providing a “front door” to preventative healthcare and access to specialty care [11]. The ENCOUNTER strategy combines a novel cost-effective, community-tailored primary care model with a geographic optimization strategy that locates care where people can use it. Preliminary work to define a scalable care model is underway: since 2015, the Rush College of Nursing (CON) has partnered with community-based organizations (CBO) to evolve a nurse-run, community-embedded primary care service model; it currently runs 32 such programs and clinics within the Chicago metropolitan area [5, 6]. The Rush model is based on the well-documented Cuban model, a decentralized network of micro community-embedded clinics anchored by a single physician and nurse to serve every 1200 residents [4]. The Cuban model is widely cited as an effective public health strategy that achieved full population coverage within 20 years of launch, and has dramatically transformed the health of residents since its deployment in the 1970s [2]. In February of 2021, the Institute of Design partnered with Rush University and its CBOs to study, formalize and systematize this place-based practice model and how each adapts services to the Chicago community it serves.

The ambition of this partnership is broad: to formalize, scale and replicate the model throughout Chicago’s South and West sides; and in the process develop a place-based public health strategy on par with Daniel Burnham’s 1909 vision (the “Burnham Plan”) that has guided the development of Chicago’s public land for the good and access of all Chicago residents for over 100 years.

Main problem. A central component of this reimagined infrastructure for primary care in Chicago is an optimization strategy to identify the ideal number and distribution of clinic locations. In the Cuban model, community based clinics are geographically distributed such that each doctor-nurse team serves up to 1200 people or approximately 300 families. By comparison, the average patient panel for a US primary care provider is 2300, a number calibrated for productivity and revenue but widely cited as contributing to access and quality issues [9]. Deriving the number and placement of clinics for Chicago (2022 metropolitan population estimated at 8,901,000) requires incorporating multiple factors to produce a real-world strategy. These factors include neighborhood population density; geographic features (highways, waterways, industrial zones); and other urban features known to drive or inhibit pedestrian traffic (street patterns, public transit, shopping districts).

A competitive external grant application would require us to do the prework of building the structure of a computational optimization model and the data it needs, so that we have precise data-supported mathematical evidence for what an ideal ENCOUNTER network should look like, and how it can be built to be a real-life flexible network that is responsive to the changing requirements of the community and the medical partners in building this health system.

Data-driven Computational Optimization Approach. Building a solution strategy for the Chicago ENCOUNTER problem will require combining facility allocation problem models with the design principles for providing equitable access to all parts of the society. Our starting framework will be a hierarchical facility location optimization model [3], where the bottom layer are the frontline doctor-nurse teams in direct contact with around 1200 patients each, and the layer above that are the community-level clinics that oversee several such doctor-nurse teams. However such a

standard optimization model is not meaningful for this problem without two fundamental modifications which will be unique to our approach. One that makes the locations for both layers equitably accessible to all segments of Chicago population. (Directly relevant is a previous project of the PI on a computational model for reallocation of public transportation resources to make existing primary care health clinics equitably accessible to all neighborhoods [10]). Second, the model has to be integrated with computational tools built and adapted to the data-based idiosyncrasies of primary healthcare need and demand in the Chicago region. This will require cleaning up and using the Census data at block and census tract level, the Chicago neighborhood and boundary data, the Chicago primary health center location and utilization data, and the Transportation data from the CTA [1]. This data would be used for the aforementioned optimization model with the programming implementation of large scale computational algorithms, e.g. a hybrid Tabu Search-Simulated Annealing algorithm that is proven to work on such problems [10].

Significance and Broad Impact. The ENCOUNTER care and optimization models are significant at three levels. At the IIT level, there is little collaboration between the Institute of Design and Applied Mathematics on the hot topic of healthcare, despite complementary skill sets and prior work. We are particularly excited for an opportunity to team our graduate students and lay the groundwork for more extensive partnerships between department faculty. At the regional level: Post-pandemic, the City of Chicago and south and westside healthcare systems are primed for a new comprehensive effort to improve the health of residents. COVID highlighted (again) that Chicago's healthcare infrastructure is driving inequity and harming the city's capacity for vitality and growth. The pandemic also upended payment models and approaches to care delivery, causing new thinking and receptivity among health system leaders about future strategies. In July, the Mayor's office released its *We Will Chicago*, a first-of-its-kind, 10-year city-wide planning framework to enhance equity and resiliency: community-based healthcare is a top priority [12]. UChicago Medicine (Inst. Design MOU pending 9/22) is actively investing in community-based care capacities; Rush University (Inst. Design MOU signed 2/21) has received a \$10m gift and hired new leadership for its Institute for Health Equity. The ENCOUNTER project will leverage these existing relationships to shape and accelerate an implementation pathway by bringing a new vision and precise tool to health systems and city leaders when evaluating public health investments and funding priorities. Lastly, ENCOUNTER is of potential significance to national public health efforts, offering a tool tailorable for use in other locales and states that face similar challenges in promoting equitable access to healthcare, including rural areas.

A PhD student from Applied Math, Alaittin Kirtisoglu, will work on developing the network optimization model for facility location that incorporates means for equitable accessibility to clinic locations and is flexible with regards to real-life requirements of our partners in Rush University, followed by writing the programming code for actual implementation of this model that utilizes the city data as discussed earlier. Alaittin spent the summer 2022 as an intern at Argonne National Lab working on a computational optimization model and its implementation, giving him the appropriate experience and expertise.

References

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